Prioritization of Quality Requirements in Decentralized Identity Applications: What Matters Most to Verifiers?

Welcome, and thank you for taking part in this survey, which is a central part of a Master Thesis project conducted at the Communication Systems Group (CSG) within the Department of Informatics at the University of Zurich.

What is this survey about?

This Master Thesis aims to prioritize the "quality requirements" for modern digital identity systems, specifically those based on Decentralized Identity (DI) and Self-Sovereign Identity (SSI). In software development, we often distinguish between "what" a system does (Functional Requirements) and how it accomplishes these tasks. These "how" aspects - usually referred to as Non-Functional Requirements (NFRs) or quality requirements - describe abstract yet critical qualities, such as consent, security, authenticity, and privacy.

Specifically, it focuses on systems built on two key concepts: Self-Sovereign Identity (SSI) and Decentralized Identity (DI).

What is Self-Sovereign Identity (SSI)?

Imagine holders using a digital wallet that contains their various forms of identification (like driver’s licenses, university degrees, or employee badges) securely on their device. SSI is a model designed to give holders complete control over that digital identity. The core goals are to enhance holders’ control over their data, guarantee its privacy, and improve trust in digital interactions. Instead of relying on a single central provider, holders decide which information to share and with whom.

What is Decentralized Identity (DI)?

DI is the underlying technology that enables SSI. Traditionally, digital identities are stored in centralized systems managed by companies or governments. A decentralized approach means there is no single point of control or failure. This architecture involves three key roles:

- **Issuers** who create and issue digital credentials
- **Holders** who store and present them
- **Verifiers** who check them

Emerging SSI/DI systems aim to give holders more control over their data, enhance privacy, and foster trust in digital interactions. This research focuses on translating these abstract goals into practical terms and understanding which qualities matter most to each role.

To do this, we are gathering feedback from the three leading roles within a decentralized identity ecosystem:

- **Holders**: Individuals who own and control their identity data.
- **Issuers**: Organizations (like universities, governments, or businesses) that provide credentials.
- **Verifiers**: Organizations (like employers or service providers) that request and verify credentials.

What will you be asked to do?

This questionnaire is designed to understand **verifiers'** priorities regarding the quality requirements of Decentralized Identity systems and should take **about 10 minutes** to complete.

The survey consists of two main parts:

1. You will be asked to rate the importance of several quality requirements (e.g., privacy, security).
2. You will be asked to choose the most and least important qualities from a given list.

There are no right or wrong answers. We are interested in your professional judgment as a **verifier**. Your participation is highly valued and will contribute significantly to this research.
There are 46 questions in this survey.

Participant background information

To help us interpret the survey results correctly, we first ask a few brief questions about your background – such as gender, year of birth, professional role, and any prior experience with Self-Sovereign Identity (SSI) or Decentralized Identity (DI).

Privacy & Confidentiality

This survey is conducted as part of a Master Thesis at the University of Zurich, and all responses are recorded fully anonymously.

- No personal identifiers: Neither names nor contact details are requested.
- **Anonymous analysis:** All data are stored and reported in aggregate form, preventing any individual identification.
- **Legal basis:** Data collection complies with the cantonal Data Protection Act (IDG) and the UZH Policy on Ethical Review of Research Projects Involving Human Subjects.
- **Voluntary participation:** You may withdraw or discard your responses at any time before submitting the survey.

Answering these items is voluntary, anonymous, and will be used **only** to analyse how perceptions of non-functional requirements may vary across different participant groups.

Please indicate your gender.

Please choose **only one** of the following:

- ◯ Female
- ◯ Male

Please enter your year of birth (YYYY).

Please write your answer here:

Which of the following best describes your current occupation or position? (Select all that apply.) *

Select all that apply

Please choose **all** that apply:

- [ ] In training (Student / Apprentice)
- [ ] Manager / Executive
- [ ] Professional / Academic occupation
- [ ] Technician and equivalent occupations
- [ ] Clerical and administrative occupations
- [ ] Service and sales worker
- [ ] Skilled worker in agriculture and forestry
- [ ] Craft and related trades
- [ ] Plant and machine operator / Assembler
- [ ] General laborer

If you are representing a company or institution in your role as verifier, please feel free to specify its name below. Otherwise, you may leave this blank to remain anonymous.

Please write your answer here:

If you wish, please describe your professional role or position (e.g., “IT Security Manager”, “Software Engineer”). You may leave this blank if you prefer not to say.

Please write your answer here:

Background

Self-Sovereign Identity, or SSI, is a new approach that puts **you** in control of your own digital identity. SSI is an approach to digital identity in which individuals (or organisations) create, store and present their own verifiable credentials without relying on a single central provider. The main idea is that your identity should belong to you, just like your physical passport or driver's license. Instead of logging in through a company, you would manage your identity from a secure personal app on your phone, often called a digital wallet.

A Decentralized Identity (DI) is your digital identity that you own and control entirely. Unlike an email address or username, which are issued and managed by specific companies, a DI is not tied to any single provider and cannot be revoked by anyone else. With your DI, you can prove who you are and share verified information from your digital wallet in a way that is secure, private, and entirely under your authority, giving you actual ownership of your digital self.

Have you ever used, implemented, or otherwise worked with SSI or DI?

*

Please choose **only one** of the following:

- ◯ Yes
- ◯ No

Question Group A – Key Qualities of Credential Verification

On the next pages you will see short statements about **important qualities in verifying digital credentials** (for example, “Credentials can always be verified by verifiers”). For each statement, please tell us how strongly you agree or disagree on a 1-to-5 scale:

- 1 = Strongly disagree 5 = Strongly agree

We are interested in your professional perspective as a **verifier** on how important these core qualities are.

Before using someone’s identity data, I want to receive clear consent from the holder every time.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Processing identity data without explicit consent from the holder would be fine for my organisation.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

When I check someone's identity, I want to see only the data that are strictly needed for my service.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

Requesting more personal data than necessary would be acceptable for our service.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need to confirm on my own that any credential I receive is genuine and still valid.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I trust credentials regardless of whether I can verify their validity myself.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Question Group B – Additional Qualities of Credential Verification

You will now see another set of statements describing **other possible qualities** of verifying digital credentials (for example, ease of checking revocation status or retrieving issuer details). Please follow the same procedure:

1. Read each statement.
2. Use the 1-to-5 scale to show how strongly you agree or disagree.

Answer every item according to your professional experience or expectations as a **verifier**. Your responses will help us understand which additional qualities verifiers value most in future digital-

identity systems.

I must be able to prove that every credential I check is genuine and unaltered.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I could accept credentials without verifying their authenticity.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need our service to accept and check credentials that come in different formats or protocols.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

Rejecting a credential just because it was issued under another standard would be manageable for us.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need to verify credentials that come from many different platforms or blockchains.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

Accepting credentials from only one ecosystem is sufficient for our service.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need to verify identity data over a channel that keeps it safe from misuse or tampering.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I would maintain trust in the system even if identity data could be intercepted or altered during verification.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I prefer to accept only credentials that follow recognized standards and schemas.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I would be fine accepting credentials that fail to meet the required standards and schemas.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need to see a clear, unalterable record of how and when credentials are verified.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

An audit trail of credential checks feels nonessential to me.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to access issuer information instantly whenever I verify a credential.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I can verify a credential effectively without retrieving issuer details.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need to verify credentials without incurring high transaction or storage fees.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I am comfortable paying high costs to verify credentials.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to verify credentials without relying on any single service.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I am comfortable relying on a single provider to check credentials.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I need confidence that security threats to identity data are effectively managed during verification.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

Effective security management during verification feels optional to me.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Priorities: What matters most — what matters least

Prioritisation of Non-Functional Requirements (NFR) in Self-Sovereign Identity (SSI) and Decentraized Identity (DI)

Over the next few pages you will complete **13 short Best–Worst choice tasks**.
Each task presents **four qualities (non-functional requirements, NFRs)** that can shape the **credential-verification process** in a decentralized-identity ecosystem—for example, Security, Cost, Privacy or Interoperability.

For every set of four:

- **Select the single quality that, in your view, is *most important*.**
- **Select a different quality that is *least important*.**

You must choose one “Most important” and one “Least important” on every screen; the same item cannot be both. There are no right or wrong answers, and the full section takes about 2 to 3 minutes.

Description

Accessibility: *Access and retrieve issuer data instantly.*

Authenticity: *Check that identity data is trustworthy and provable.*

Compatibility: *Import and validate credentials under various standards and protocols.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Accessibility	○	○
Authenticity	○	○
Compatibility	○	○

Description

Consent: *Request only minimal data and ensure GDPR-compliant consent.*

Cost: *Minimize implementation and operational costs.*

Decentralization: *Resolve DIDs via a distributed registry, avoiding central infrastructure.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Consent	○	○
Cost	○	○
Decentralization	○	○

Description

Interoperability: *Verify credentials across diverse platforms and services.*

Privacy: *Process only the data necessary for the service.*

Protection: *Protect identity data from misuse during verification.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Interoperability	○	○
Privacy	○	○
Protection	○	○

Description

Security: *Authenticate holders and secure data before trust decisions.*

Standard: *Require adherence to specific schemas or trusted issuers.*

Transparency: *Operate transparent validation processes.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Security	○	○
Standard	○	○
Transparency	○	○

Description

Verifiability: *Independently verify credential validity and authenticity.*

Accessibility: *Access and retrieve issuer data instantly.*

Authenticity: *Check that identity data is trustworthy and provable.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Verifiability	○	○
Accessibility	○	○
Authenticity	○	○

Description

Compatibility: *Import and validate credentials under various standards and protocols.*

Consent: *Request only minimal data and ensure GDPR-compliant consent.*

Cost: *Minimize implementation and operational costs.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Compatibility	○	○
Consent	○	○
Cost	○	○

Description

Decentralization: *Resolve DIDs via a distributed registry, avoiding central infrastructure.*

Interoperability: *Verify credentials across diverse platforms and services.*

Privacy: *Process only the data necessary for the service.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Decentralization	○	○
Interoperability	○	○
Privacy	○	○

Description

Protection: *Protect identity data from misuse during verification.*

Security: *Authenticate holders and secure data before trust decisions.*

Standard: *Require adherence to specific schemas or trusted issuers.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Protection	○	○
Security	○	○
Standard	○	○

Description

Transparency: *Operate transparent validation processes.*

Verifiability: *Independently verify credential validity and authenticity.*

Accessibility: *Access and retrieve issuer data instantly.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Transparency	○	○
Verifiability	○	○
Accessibility	○	○

Description

Authenticity: *Check that identity data is trustworthy and provable.*

Compatibility: *Import and validate credentials under various standards and protocols.*

Consent: *Request only minimal data and ensure GDPR-compliant consent.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Authenticity	○	○
Compatibility	○	○
Consent	○	○

Description

Cost: *Minimize implementation and operational costs.*

Decentralization: *Resolve DIDs via a distributed registry, avoiding central infrastructure.*

Interoperability: *Verify credentials across diverse platforms and services.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Cost	○	○
Decentralization	○	○
Interoperability	○	○

Description

Privacy: *Process only the data necessary for the service.*

Protection: *Protect identity data from misuse during verification.*

Security: *Authenticate holders and secure data before trust decisions.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Privacy	◯	◯
Protection	◯	◯
Security	◯	◯

Description

Standard: *Require adherence to specific schemas or trusted issuers.*

Transparency: *Operate transparent validation processes.*

Verifiability: *Independently verify credential validity and authenticity.*

*

Please select exactly one 'Most important' and one 'Least important' option.
Please choose the appropriate response for each item:

	Most important	Least important
Standard	○	○
Transparency	○	○
Verifiability	○	○

You have successfully completed the survey.

Thank you very much for your valuable time and participation. With your detailed answers, you have made a crucial contribution to my Master's thesis, which focuses on the prioritization of

software quality requirements for future digital identity systems (Self-Sovereign Identity). Your perspective is essential for this research.

If you have any further questions, comments or remarks about this survey, please feel free to enter them in the text field below.

Please write your answer here:

Thank you for your time and valuable contribution to this research.

The data you have provided will be analyzed to understand how different groups prioritize qualities in Decentralized Identity systems.

Your input is essential for fulfilling the goals of this Master Thesis. We appreciate your willingness to share your perspective.

Submit your survey.
Thank you for completing this survey.